

TaskSync – Collaborative Workspace

Project Development Approach Document

Project Overview

TaskSync is a collaborative project management platform designed to help teams manage projects and tasks efficiently while ensuring secure collaboration between multiple users. Unlike a traditional to-do application, TaskSync focuses on teamwork by allowing project owners to create projects, invite registered users, assign responsibilities, track progress, and monitor project activities through a centralized workspace.

The objective of this project is to build a secure, modern, and user-friendly collaborative workspace using **React + Vite** for the frontend, **Node.js and Express.js** for the backend, and **PostgreSQL** as the database. Along with implementing all the required functionalities, the focus will be on writing clean, modular, and maintainable code while ensuring a smooth user experience.

Development Philosophy

My approach is to develop the application module by module instead of building all the features together. Each module will be designed, implemented, tested, and verified before moving to the next stage. This approach will make debugging easier, improve code quality, and ensure that every part of the application works correctly before integrating it with the complete system.

The development will follow this sequence:

Database Design → Backend APIs → Security & Authorization → React Frontend Development → Interactive Features → Testing and Bug Fixing

Technology Stack

Frontend

- React
- Vite
- HTML5
- CSS3
- JavaScript (ES6+)

Backend

- Node.js
- Express.js
- bcrypt
- JSON Web Token (JWT)

Database

- PostgreSQL
-

Development Approach

Phase 1 – Database Design

The first phase of the project will focus on designing the PostgreSQL database according to the project requirements.

The database will contain the following tables:

- Users
- Projects
- Project Members
- Tasks

Additional tables will also be created to support enhanced functionality:

- Activity Logs
- Task History

The relationships between these tables will be established using foreign keys to maintain data consistency and ensure secure access to project information.

Once the database structure is completed, all database operations will be tested before proceeding with backend development.

Phase 2 – Authentication System

After completing the database, the authentication system will be implemented.

This module will include:

- User Registration
- User Login

- Password Encryption using bcrypt
- JWT Authentication
- HttpOnly Cookie-based Authentication
- Logout Functionality

Only authenticated users will be allowed to access the application.

Phase 3 – Authorization & Security

Security is one of the most important aspects of this project.

Every protected API request will verify:

- Whether the user is authenticated.
- Whether the requested project exists.
- Whether the user is a member of that project.
- Whether the user has the required permission (Admin or Member).

This authorization process will ensure that users can only access projects they are invited to and cannot view or modify data belonging to other projects.

Reusable authorization middleware will be implemented to keep the backend organized and maintainable.

Phase 4 – Project Management Module

Once authentication and authorization are completed, the project management module will be developed.

The module will provide the following features:

- Create Project
- View Accessible Projects
- Invite Team Members
- Manage Project Members
- Role-Based Access (Admin / Member)

Project creation and member assignment will be implemented using PostgreSQL Transactions to maintain database consistency.

Phase 5 – Task Management Module

Each project will contain multiple tasks.

The task management module will provide the following features:

- Create Task
- Assign Task
- Edit Task
- Update Task Status
- Move Tasks Across Workflow Stages

Before any task is modified, the backend will verify both project membership and user permissions.

Phase 6 – React Frontend Development

The frontend will be developed using **React** with **Vite** to create a fast and responsive Single Page Application.

The application interface will be divided into reusable components, making the code easier to understand, maintain, and extend in the future.

The frontend will include the following major screens:

- Login & Registration
- Dashboard
- Project Workspace
- Kanban Board

Navigation between different pages will be smooth without requiring a full page reload, providing a better user experience.

Phase 7 – Interactive Kanban Board

Each project will display a Kanban Board containing three workflow stages:

- To Do
- In Progress
- Done

Tasks will automatically appear in their respective columns.

Whenever a task status changes, the frontend will immediately reflect the updated information after receiving a successful response from the backend.

Phase 8 – Testing and Bug Fixing

The final phase will focus on ensuring the application is stable and reliable.

This phase will include:

- API Testing
- Database Validation
- Error Handling
- Security Testing
- Responsive Design Testing
- Bug Fixing
- Code Refactoring

Every module will be tested independently before integrating it into the final application.

Mandatory Bonus Features

The following bonus features mentioned in the project requirements will also be implemented.

1. Drag and Drop Kanban Board

The Kanban Board will support native HTML5 Drag and Drop functionality.

Users will be able to drag a task from one workflow column to another.

Once dropped:

- The frontend will immediately update.
- The backend will update the task status.
- The activity log will record the change automatically.

This will provide a more interactive and efficient task management experience.

2. Activity Feed

An Activity Logs module will be developed.

The application will record important project activities such as:

- Project Invitations
- Task Creation
- Task Assignment

- Task Status Changes

A Recent Activity section will display these updates so that team members can easily track project progress.

3. Project Progress Calculation

The dashboard will display the completion percentage for every project.

The backend will calculate this percentage by comparing the number of completed tasks with the total number of tasks in the project, and the frontend will display the result using a progress bar.

Additional Features

Frontend Feature – Advanced Task Filtering and Search

To improve task management, users will be able to:

- Search tasks by title.
- Filter tasks by status.
- Filter tasks by assigned member.

The displayed tasks will update instantly based on the selected filters, making it easier to manage larger projects.

Backend Feature – Task History

In addition to the activity feed, the application will maintain a detailed history of task updates.

Whenever a task is modified, the following information will be stored:

- Previous Value
- Updated Value
- User Who Made the Change
- Date and Time

This feature will help track modifications made throughout the project's lifecycle.

Additional Feature – Intelligent Workload Balancer

To improve task distribution, the application will calculate the number of active tasks assigned to each project member.

While assigning a task, the system will recommend the member with the lowest current workload.

This feature will encourage balanced work distribution and improve overall team productivity.

Expected Outcome

The completed project will be a secure, modern, and user-friendly collaborative workspace where multiple users can work together efficiently on shared projects.

The application will demonstrate practical implementation of:

- React-based Single Page Application (SPA)
- Component-Based Frontend Development
- Authentication and Authorization
- Relational Database Design
- SQL JOIN Operations
- PostgreSQL Transactions
- RESTful API Development
- Interactive Kanban Board
- Drag-and-Drop Functionality
- Activity Tracking
- Task History
- Project Progress Monitoring
- Secure Multi-user Collaboration

Overall, this project aims to simulate a real-world project management platform by following clean coding practices, modular development, and industry-standard software engineering principles.